

Filipe Alves Cavalcante

Senior Full-Stack Software Engineer

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SUMMARY

Senior full-stack engineer with nearly 8 years of experience, running my own software consultancy since 2018. I've delivered for large-scale edtech (a platform that grew from 300,000 to 800,000 paying students), a multi-tenant ERP and B2B marketplace, one of Brazil's largest media groups and a portfolio of agency clients — often as the sole developer, owning database design, back-end, front-end, performance and deployment end to end.

SKILLS

Languages: PHP, JavaScript, TypeScript, Go, SQL

Frameworks: Laravel, NestJS, Node.js, React, Next.js, Vue.js, Express, Fastify, React Native, Tailwind CSS

Architecture: REST APIs, Microservices, Multi-tenant SaaS, Event-Driven, CQRS, Domain-Driven Design, System Design, Scalability

Cloud & DevOps: AWS (Lambda, EC2, S3, SQS, DynamoDB), Docker, CI/CD, Linux, Apache, Nginx

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis — query optimization, indexing, performance tuning

AI & Integrations: OpenAI Whisper, AI-assisted development (Claude Code, Codex), Mercado Livre API, Google Ads API, Instagram Graph API, TikTok API

Testing: PHPUnit, Jest, Pest, TDD, Unit Testing, Integration Testing

Practices: Agile/Scrum, Code Review, Mentoring, Performance Tuning, Observability, Security, Accessibility (WCAG), Production Support

EXPERIENCE

Senior Full-Stack Software Engineer

Jul 2025 – Present

BrasilTec | Remote · contract via F A Cavalcante (my consultancy)

Multi-tenant ERP and B2B marketplace (Mercado Livre, Google Ads, virtual storefront) plus the company's online university platform — 6 tenant databases. Sole developer on the university platform; alongside 3 senior and 2 junior engineers on the ERP.

- Rebuilt the university platform from scratch in **6 weeks** — back-end, front-end, admin panel and data migration — replacing a legacy PHP system with NestJS + React/TypeScript. It now serves **5,143 users and 6,983 enrollments**.
- Cut the application server footprint by **50%+** by moving product photos, course videos and generated captions off the box to **Amazon S3** and a CDN, replacing a setup where every asset lived on the server itself.
- Made every lesson usable by deaf and hard-of-hearing students and by the **10% of learners outside Brazil**, with an automated captioning pipeline (OpenAI Whisper) that generates Portuguese, English and Spanish subtitles across **100+ hours of video**.
- Cut the heaviest product query from **1.21s to 47ms (26×)** and up to **99.7% of database round-trips** by eliminating N+1 patterns across a 6-tenant PostgreSQL system, validated with **zero divergence across 65,000 products**. Tuning shared_buffers, JIT thresholds and functional indexes took cache hit ratio to **98.92%** and the worst polling query from **653ms to 1.9ms**.
- Rewrote a legacy PHP playlist query from **470ms to 0.18ms** and closed a **SQL injection** in the same file, deployed with no schema change and a one-command rollback path.
- Recovered **82GB of disk (91% to 81%)** and eliminated **~3% network packet loss** on the production host running 147 containers — zero service interruption, fixes persisted across reboots via systemd.
- Turned the **entire student base** into qualified leads inside the company's sales platform, unifying legacy and new enrollment records through platform-to-platform ERP integration.
- Shipped AI and content features to production: lesson search over transcripts, auto-generated summaries with exact video timestamps, an AI question generator, and a tool that cuts long videos into reels and schedules posts to Instagram and TikTok.

Senior Software Engineer

Dec 2021 – Jun 2025

Gran Cursos Online | Brasília, Brazil (Remote) · full-time

Large-scale edtech platform running Scrum. The company went from 300,000 to 800,000 paying students while I was there.

- Led a MySQL to PostgreSQL migration with query rewrites and index tuning that saved **\$1,000+/month** in infrastructure.
- Shipped the Black Friday campaign in **5 days against a 10-day deadline** — re-themed the whole platform to a black theme with new banners and promo pages for the university and post-graduation sites. It beat its **R\$10M first-day target and closed the day at R\$16M**.
- Built that re-skin as a theming layer instead of one-off pages, so marketing and design could keep changing banners and landing pages without engineering.
- Published a shared React component library on NPM adopted across **4 repositories**, covering 20+ components and removing thousands of lines of duplicated code — reviewing the teams' pull requests as they migrated.
- Rebuilt the company marketing site for performance and SEO — Lighthouse Performance **7 to 76** and SEO **28 to 95**, then worked through the Google Search Console task backlog to lift the site's Google rankings.

- Removed engineering from the page-publishing path with a CMS the marketing team runs themselves, and closed customer-journey tracking gaps by instrumenting the platform with **Amplitude** and a **Google Tag Manager dataLayer** for marketing and product stakeholders.
- Ended a question-ordering bug that had resurfaced for years — the cause wasn't in the code but an undocumented business rule, which I traced and resolved cross-functionally with the team that owned the question registration system.

Software Engineer

Jul 2021 – Dec 2021

Grupo Estado | São Paulo, Brazil (Remote) · contract via F A Cavalcante

One of Brazil's largest media groups. Part of the production support team, handling maintenance and improvements across multiple news platforms.

- Cut Google Search Console errors by **more than 50%** on a platform carrying 20,000+ daily errors, by adding structured logging, tracing where they came from and fixing the code issues one by one.
- Removed manual post-release payment checks entirely by automating a bot that tested payments on every deploy and alerted the right teams on Email, Slack and Hangouts with screenshots and error details.
- Stopped **500+ broken links per month** across 3 news platforms with a content indexing and monitoring system in Node.js and TypeScript using Google APIs.
- Backed the **L1 and L2 support teams** directly: reviewed their fixes and walked them through diagnosing recurring failures so more issues were closed without reaching engineering.

Software Engineer

Jul 2020 – Jul 2021

Olivas Digital | São Paulo, Brazil (Remote) · contract via F A Cavalcante

Marketing agency with 13 clients in the portfolio. Full-stack across all of them using PHP, Laravel, Node.js, Express, React, Vue or React Native depending on the project.

- Made releases predictable across the agency's projects by replacing FTP deploys with CI/CD pipelines — before that, deploys meant wrong files, broken states and no rollback, and the team dreaded them.
- Generated **4 new clients through referrals** with landing pages built for several accounts.
- Integrated anti-fraud APIs for a charity receiving donations to rescue trafficked animals — chargebacks dropped **40%**, the best fraud result across all agency clients.
- Gave the client visibility into international sales for the first time with a multi-currency REST API covering 5 currencies, order tracking and commission reporting.

Software Engineer

Dec 2018 – Jun 2020

Gênesis Tecnologia e Inovação | São Paulo, Brazil (Remote) · contract via F A Cavalcante

- Replaced a legacy enterprise system split between Java and PHP 5.6 with a Laravel rewrite — no documentation, no handover: I read the old code, redesigned the database and rebuilt the back-end from scratch.
- Swapped a manual spreadsheet reporting process for backend-generated reports — overnight batch to on demand, **50% faster**.

PROJECTS

TEM AQUI Achadinhos — temaquiachadinhos.com.br

2026 – Present

Live in production · Go · chi · pgx + sqlc · goose · PostgreSQL 17 · Next.js 16 · TypeScript · Railway

- Affiliate storefront with a self-hosted link shortener on my own domain: every click is counted and redirected preserving the exact affiliate URL.
- **Go** back-end (chi, pgx + sqlc, embedded goose migrations, argon2id, LRU cache, structured logging) served behind a single public Next.js origin — one origin means no CORS, same-origin cookies and a simple CSP.

Lembru

2026

Laravel 13 + FrankenPHP · React + Vite · React Native (Expo) · Keycloak · PostgreSQL · Redis · Caddy · AWS SES

- Multi-container platform behind a Caddy gateway — React web, Laravel API, queue worker, scheduler, Redis and PgBouncer-pooled PostgreSQL — plus a React Native app with biometric auth and push notifications.
- Separated identity from application data by design — **Keycloak OIDC** with Authorization Code + PKCE, an identity database the API cannot reach, and token validation against the provider's public keys.
- Self-hosted transactional email on **Amazon SES** instead of a paid provider, keeping delivery at SES per-message cost.

EDUCATION

Coursework in System Analysis and Development — Instituto Federal de São Paulo

2016 – 2018

IT Technician — Senac, São Paulo

2013 – 2015

CERTIFICATIONS

Clean Code and Clean Architecture — Branas (2024) · GoStack — Rocketseat (2020)

LANGUAGES

Portuguese (Native) · English (B2)